

Lagged Covariances

under Piecewise-Constant Means

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1 Setup and notation

Let $X_1, \dots, X_n \in \mathbb{R}^p$ with

$$X_t = \theta_t + \varepsilon_t,$$

where $\theta_t \in \mathbb{R}^p$ is a deterministic, piecewise-constant mean with finitely many change points, and the noise ε_t is i.i.d. across t with $\mathbb{E}[\varepsilon_t] = 0$ and $\text{Cov}(\varepsilon_t) = \Sigma$. We write $\varepsilon_{t,a}$, $\theta_{t,a}$, $X_{t,a}$ for the a -th coordinate at time t . Because the noise is i.i.d. across time, $\mathbb{E}[\varepsilon_{u,i}\varepsilon_{v,i}] = \sigma_i^2 \mathbf{1}\{u = v\}$. We call a pair of consecutive times $(\tau, \tau + 1)$ a **gap**, and two gaps *adjacent* when they share a time. A **change point** is a gap at which the mean vector θ changes: at least one coordinate moves there.

Assumption 1 (Null model). $\Sigma = \text{diag}(\sigma_1^2, \dots, \sigma_p^2)$. The coordinates of ε_t are mutually independent. Moreover, any three consecutive time points have at most one change point between them. Equivalently, no two change points fall in adjacent gaps.

Remark. Assumption 1 yields **cross-coordinate independence**, so $\mathbb{E}[\varepsilon_{u,i}\varepsilon_{v,j}] = \mathbb{E}[\varepsilon_{u,i}\varepsilon_{v,j}\varepsilon_{w,k}] = \mathbb{E}[\varepsilon_{u,i}\varepsilon_{v,j}\varepsilon_{w,k}\varepsilon_{x,l}] = 0$ whenever one coordinate index differs from all the others.

We call a difference of the noise at two time points, for a fixed coordinate, a **noise increment**: the noise increment in coordinate i over the time pair (α, α') is $\varepsilon_{\alpha,i} - \varepsilon_{\alpha',i}$. Likewise, $\theta_{\alpha,i} - \theta_{\alpha',i}$ is the **mean jump** in coordinate i over that time pair. For the mean jump between two consecutive times τ and $\tau + 1$ we write $\delta_i^\tau := \theta_{\tau,i} - \theta_{\tau+1,i}$.

$$\begin{aligned} \widehat{s}_{t,(a,b)} &:= \frac{1}{2}(X_{t,a} - X_{t+1,a})(X_{t,b} - X_{t+1,b}) \\ &\quad + \frac{1}{2}(X_{t+1,a} - X_{t+2,a})(X_{t+1,b} - X_{t+2,b}) \\ &\quad - \frac{1}{2}(X_{t,a} - X_{t+2,a})(X_{t,b} - X_{t+2,b}). \end{aligned} \tag{1}$$

We call the three consecutive times $\{t, t + 1, t + 2\}$ on which $\widehat{s}_{t,(a,b)}$ is built its **window**.

2 The $Q + L + C$ decomposition

Insert $X_{u,i} = \theta_{u,i} + \varepsilon_{u,i}$ into (1) and collect terms by the number of noise factors they carry. This gives $\widehat{s}_{t,(a,b)} = Q_{t,(a,b)} + L_{t,(a,b)} + C_{t,(a,b)}$, with the quadratic part

$$\begin{aligned} Q_{t,(a,b)} &= \frac{1}{2}(\varepsilon_{t,a} - \varepsilon_{t+1,a})(\varepsilon_{t,b} - \varepsilon_{t+1,b}) \\ &\quad + \frac{1}{2}(\varepsilon_{t+1,a} - \varepsilon_{t+2,a})(\varepsilon_{t+1,b} - \varepsilon_{t+2,b}) \\ &\quad - \frac{1}{2}(\varepsilon_{t,a} - \varepsilon_{t+2,a})(\varepsilon_{t,b} - \varepsilon_{t+2,b}), \end{aligned} \tag{2}$$

the linear part

$$\begin{aligned} L_{t,(a,b)} &= \frac{1}{2}(\theta_{t,a} - \theta_{t+1,a})(\varepsilon_{t,b} - \varepsilon_{t+1,b}) + \frac{1}{2}(\varepsilon_{t,a} - \varepsilon_{t+1,a})(\theta_{t,b} - \theta_{t+1,b}) \\ &\quad + \frac{1}{2}(\theta_{t+1,a} - \theta_{t+2,a})(\varepsilon_{t+1,b} - \varepsilon_{t+2,b}) + \frac{1}{2}(\varepsilon_{t+1,a} - \varepsilon_{t+2,a})(\theta_{t+1,b} - \theta_{t+2,b}) \\ &\quad - \frac{1}{2}(\theta_{t,a} - \theta_{t+2,a})(\varepsilon_{t,b} - \varepsilon_{t+2,b}) - \frac{1}{2}(\varepsilon_{t,a} - \varepsilon_{t+2,a})(\theta_{t,b} - \theta_{t+2,b}), \end{aligned} \tag{3}$$

and the constant part

$$\begin{aligned} C_{t,(a,b)} &= \frac{1}{2}(\theta_{t,a} - \theta_{t+1,a})(\theta_{t,b} - \theta_{t+1,b}) \\ &\quad + \frac{1}{2}(\theta_{t+1,a} - \theta_{t+2,a})(\theta_{t+1,b} - \theta_{t+2,b}) \\ &\quad - \frac{1}{2}(\theta_{t,a} - \theta_{t+2,a})(\theta_{t,b} - \theta_{t+2,b}). \end{aligned} \tag{4}$$

Note. By Assumption 1 the window $\{t, t+1, t+2\}$ holds at most one change point, so $C_{t,(a,b)} = 0$. This is what makes $\hat{s}_{t,(a,b)}$ unbiased for $\sigma_{a,b}$.

Convention and symmetry. We label a pair by its two distinct coordinates in increasing order, so (a, b) always means $a < b$, and $\hat{s}_{t,(a,b)}$ is defined only for such pairs. The two coordinates enter (1) through the same expression, and interchanging them leaves each product unchanged. A pair is therefore effectively an unordered set $\{a, b\}$, and every quantity built from $\hat{s}_{t,(a,b)}$ below depends only on that set, not on which coordinate is written first. We accordingly write each pair with its smaller index first, so $a < b$ and $c < d$.

3 Reduction of the covariance at lag m

Fix a lag $m \geq 0$. Because $C_{t,(a,b)}$ and $C_{t+m,(c,d)}$ are constants,

$$\begin{aligned} \text{Cov}(\hat{s}_{t,(a,b)}, \hat{s}_{t+m,(c,d)}) &= \text{Cov}(Q_{t,(a,b)}, Q_{t+m,(c,d)}) + \text{Cov}(Q_{t,(a,b)}, L_{t+m,(c,d)}) \\ &\quad + \text{Cov}(L_{t,(a,b)}, Q_{t+m,(c,d)}) + \text{Cov}(L_{t,(a,b)}, L_{t+m,(c,d)}). \end{aligned}$$

Proposition 1 (The two cross terms vanish at every lag). For all $m \geq 0$, $\text{Cov}(Q_{t,(a,b)}, L_{t+m,(c,d)}) = \text{Cov}(L_{t,(a,b)}, Q_{t+m,(c,d)}) = 0$.

Proof. A term of $L_{t,(a,b)} Q_{t+m,(c,d)}$ is a deterministic mean jump times $\varepsilon_{p,i} \varepsilon_{q,c} \varepsilon_{w,d}$, with one noise factor from $L_{t,(a,b)}$ carrying a coordinate $i \in \{a, b\}$ and the two factors from $Q_{t+m,(c,d)}$ carrying the coordinates $c \neq d$. The three noise coordinates form the multiset $\{i, c, d\}$. Since $c \neq d$, some coordinate occurs exactly once. By cross-coordinate independence the expectation factorizes and that lone factor contributes $\mathbb{E}[\varepsilon] = 0$. This argument counts coordinates, not times, so it is unaffected by the lag m . Hence $\mathbb{E}[L_{t,(a,b)} Q_{t+m,(c,d)}] = 0$, and since $\mathbb{E}[L_{t,(a,b)}] = 0$ the covariance vanishes. The other term is identical by symmetry. $\square$

$$\text{Cov}(\hat{s}_{t,(a,b)}, \hat{s}_{t+m,(c,d)}) = \underbrace{\text{Cov}(Q_{t,(a,b)}, Q_{t+m,(c,d)})}_{\text{quadratic part}} + \underbrace{\text{Cov}(L_{t,(a,b)}, L_{t+m,(c,d)})}_{\text{linear part}} \quad (m \geq 0).$$

4 Covariance of the quadratic parts $\text{Cov}(Q_{t,(a,b)}, Q_{t+m,(c,d)})$

Fix a pair of coordinates $a < b$. Here $\sigma_{a,b}$ is the (a, b) entry of the covariance matrix Σ . It lies off the diagonal of Σ , and under the null Σ is diagonal, so $\sigma_{a,b} = 0$, and likewise $\sigma_{c,d} = 0$ for the second pair $c < d$. Hence $\mathbb{E}[Q_{t,(a,b)}] = \mathbb{E}[Q_{t+m,(c,d)}] = 0$ and

$$\text{Cov}(Q_{t,(a,b)}, Q_{t+m,(c,d)}) = \mathbb{E}[Q_{t,(a,b)} Q_{t+m,(c,d)}].$$

$Q_{t,(a,b)}$ contains three terms. Each term is a product of two increments over the same time pair, one in coordinate a and one in coordinate b , so each term is labelled by that single time pair. The three time pairs of $Q_{t,(a,b)}$ are $(t, t+1)$, $(t+1, t+2)$, $(t, t+2)$. Write s_I for the sign of the term with time pair I , so $s_{(t,t+1)} = s_{(t+1,t+2)} = +1$ and $s_{(t,t+2)} = -1$ (the signs in (2)). By the same reading the three time pairs of $Q_{t+m,(c,d)}$ are $(t+m, t+m+1)$, $(t+m+1, t+m+2)$,

$(t+m, t+m+2)$, with the same signs s_J . Thus $Q_{t,(a,b)}$ uses the times $t, t+1, t+2$ and $Q_{t+m,(c,d)}$ uses $t+m, t+m+1, t+m+2$: the two windows coincide when $m=0$, share the times $t+1, t+2$ when $m=1$, and only the time $t+2$ when $m=2$.

The product $Q_{t,(a,b)} Q_{t+m,(c,d)}$ has $3 \times 3 = 9$ terms. We take one such term: a term of $Q_{t,(a,b)}$ whose time pair is $I = (\alpha, \alpha')$, times a term of $Q_{t+m,(c,d)}$ whose time pair is $J = (\gamma, \gamma')$. Summing over the nine choices,

$$\mathbb{E}[Q_{t,(a,b)} Q_{t+m,(c,d)}] = \frac{1}{4} \sum_{I,J} s_I s_J \mathbb{E}[(\varepsilon_{\alpha,a} - \varepsilon_{\alpha',a})(\varepsilon_{\alpha,b} - \varepsilon_{\alpha',b})(\varepsilon_{\gamma,c} - \varepsilon_{\gamma',c})(\varepsilon_{\gamma,d} - \varepsilon_{\gamma',d})].$$

■ When the two coordinate pairs differ.

Take one term. Its expectation is that of a product of four noise factors, in coordinates a, b (from the $Q_{t,(a,b)}$ side) and c, d (from the $Q_{t+m,(c,d)}$ side). Variables with different coordinates are mutually independent, so the expectation splits into a product over the distinct coordinates present, and any coordinate appearing a single time contributes a factor $\mathbb{E}[\varepsilon] = 0$. Since $a \neq b$ and $c \neq d$, every coordinate appears at least twice only when $(a, b) = (c, d)$. The expectation is therefore zero unless the two coordinate pairs coincide.

■ When the two coordinate pairs coincide.

Set $(a, b) = (c, d)$. The four factors are then two increments in coordinate a (over I and over J) and two in coordinate b (over I and over J). As a and b are independent, the expectation factorizes as

$$\mathbb{E}[(\varepsilon_{\alpha,a} - \varepsilon_{\alpha',a})(\varepsilon_{\gamma,a} - \varepsilon_{\gamma',a})] \cdot \mathbb{E}[(\varepsilon_{\alpha,b} - \varepsilon_{\alpha',b})(\varepsilon_{\gamma,b} - \varepsilon_{\gamma',b})].$$

Each factor is the covariance of the increment over I and the increment over J for the same coordinate. Using $\mathbb{E}[\varepsilon_{p,i} \varepsilon_{q,i}] = \sigma_i^2 \mathbf{1}\{p=q\}$,

$$\mathbb{E}[(\varepsilon_{\alpha,i} - \varepsilon_{\alpha',i})(\varepsilon_{\gamma,i} - \varepsilon_{\gamma',i})] = \sigma_i^2 g_{IJ}, \quad g_{IJ} := \mathbf{1}\{\alpha = \gamma\} - \mathbf{1}\{\alpha = \gamma'\} - \mathbf{1}\{\alpha' = \gamma\} + \mathbf{1}\{\alpha' = \gamma'\}.$$

In particular $g_{II} = 2$: a single increment has variance $2\sigma_i^2$. Both coordinates run over the same time pairs I and J , so each contributes the same g_{IJ} , and the term equals $\sigma_a^2 \sigma_b^2 g_{IJ}^2$. Hence

$$\mathbb{E}[Q_{t,(a,b)} Q_{t+m,(c,d)}] = \frac{1}{4} \sigma_a^2 \sigma_b^2 \sum_{I,J} s_I s_J g_{IJ}^2.$$

Lag $m=0$. When $m=0$, $Q_{t,(a,b)}$ and $Q_{t,(c,d)}$ are built on the same window, so I and J run over the same three time pairs $(t, t+1)$, $(t+1, t+2)$, $(t, t+2)$ and the increments can coincide ($g_{II} = 2$). The g_{IJ} are

g_{IJ}	$(t, t+1)$	$(t+1, t+2)$	$(t, t+2)$
$(t, t+1)$	2	-1	1
$(t+1, t+2)$	-1	2	1
$(t, t+2)$	1	1	2

Squaring and attaching the signs, the row sums are

$$\underbrace{(+4) + (+1) + (-1)}_{I=(t, t+1)} + \underbrace{(+1) + (+4) + (-1)}_{I=(t+1, t+2)} + \underbrace{(-1) + (-1) + (+4)}_{I=(t, t+2)} = 4 + 4 + 2 = 10.$$

Hence $\mathbb{E}[Q_{t,(a,b)} Q_{t,(c,d)}] = \frac{1}{4} \cdot 10 \sigma_a^2 \sigma_b^2 = \frac{5}{2} \sigma_a^2 \sigma_b^2$.

Lag $m = 1$. The time pairs of $Q_{t+1,(c,d)}$ are $(t+1, t+2)$, $(t+2, t+3)$, $(t+1, t+3)$, and the g_{IJ} are

g_{IJ}	$(t+1, t+2)$	$(t+2, t+3)$	$(t+1, t+3)$
$(t, t+1)$	-1	0	-1
$(t+1, t+2)$	2	-1	1
$(t, t+2)$	1	-1	0

Squaring each entry and attaching the signs $s_I s_J$ (both s_I and s_J equal $+1, +1, -1$), the row sums are

$$\underbrace{(+1) + (+0) + (-1)}_{I=(t, t+1)} + \underbrace{(+4) + (+1) + (-1)}_{I=(t+1, t+2)} + \underbrace{(-1) + (-1) + (+0)}_{I=(t, t+2)} = 0 + 4 - 2 = 2.$$

The single large contribution is $I = J = (t+1, t+2)$, the increment over the time pair shared by the two windows, where $g_{IJ} = 2$. Hence $\mathbb{E}[Q_{t,(a,b)} Q_{t+1,(c,d)}] = \frac{1}{4} \cdot 2 \sigma_a^2 \sigma_b^2 = \frac{1}{2} \sigma_a^2 \sigma_b^2$.

Lag $m = 2$. Now $Q_{t+2,(c,d)}$ has time pairs $(t+2, t+3)$, $(t+3, t+4)$, $(t+2, t+4)$, and the two windows share only the single time $t+2$, so no time pair of $Q_{t,(a,b)}$ can equal a time pair of $Q_{t+2,(c,d)}$. The g_{IJ} are

g_{IJ}	$(t+2, t+3)$	$(t+3, t+4)$	$(t+2, t+4)$
$(t, t+1)$	0	0	0
$(t+1, t+2)$	-1	0	-1
$(t, t+2)$	-1	0	-1

Squaring and attaching the signs, the row sums are

$$\underbrace{(+0) + (+0) + (-0)}_{I=(t, t+1)} + \underbrace{(+1) + (+0) + (-1)}_{I=(t+1, t+2)} + \underbrace{(-1) + (-0) + (+1)}_{I=(t, t+2)} = 0.$$

Hence $\mathbb{E}[Q_{t,(a,b)} Q_{t+2,(c,d)}] = 0$.

$$\text{Cov}(Q_{t,(a,b)}, Q_{t+m,(c,d)}) = \begin{cases} \frac{5}{2} \sigma_a^2 \sigma_b^2, & m = 0 \text{ and } (a, b) = (c, d), \\ \frac{1}{2} \sigma_a^2 \sigma_b^2, & m = 1 \text{ and } (a, b) = (c, d), \\ 0, & \text{otherwise.} \end{cases}$$

The quadratic part is **one-dependent**: nonzero only at lags 0 and 1, and only when the two coordinate pairs coincide.

5 Covariance of the linear parts $\text{Cov}(L_{t,(a,b)}, L_{t+m,(c,d)})$

Since $\mathbb{E}[L_{t,(a,b)}] = 0$ and $\mathbb{E}[L_{t+m,(c,d)}] = 0$, their covariance equals the product moment

$$\text{Cov}(L_{t,(a,b)}, L_{t+m,(c,d)}) = \mathbb{E}[L_{t,(a,b)} L_{t+m,(c,d)}].$$

By (3), each of $L_{t,(a,b)}$ and $L_{t+m,(c,d)}$ is a sum of six terms, so the product $L_{t,(a,b)} L_{t+m,(c,d)}$ expands into 36 terms. We examine which of these have nonzero expectation.

A single term of the product. Each such term of $L_{t,(a,b)}$ is a mean jump in one of the coordinates a, b times a noise increment in the other coordinate, both over the same time pair, and likewise for $L_{t+m,(c,d)}$ in the coordinates c, d . A term of $L_{t,(a,b)}$ is fixed by its time pair together with which of its two coordinates carries the mean jump. Take the term of $L_{t,(a,b)}$ with time pair $I = (\alpha, \alpha')$ and the mean jump in coordinate a (so the increment is in coordinate b), and the term of $L_{t+m,(c,d)}$ with time pair $J = (\gamma, \gamma')$ and the mean jump in coordinate c (so the increment is in coordinate d). Their product is

$$\frac{1}{4} s_I s_J \underbrace{(\theta_{\alpha,a} - \theta_{\alpha',a})(\theta_{\gamma,c} - \theta_{\gamma',c})}_{\text{deterministic mean jumps}} \underbrace{(\varepsilon_{\alpha,b} - \varepsilon_{\alpha',b})(\varepsilon_{\gamma,d} - \varepsilon_{\gamma',d})}_{\text{noise increments}},$$

where $s_I, s_J \in \{+1, -1\}$ are the signs attached to the time pairs I and J . Its expectation is that deterministic factor times $\mathbb{E}[(\varepsilon_{\alpha,b} - \varepsilon_{\alpha',b})(\varepsilon_{\gamma,d} - \varepsilon_{\gamma',d})]$, the covariance of two noise increments: one in coordinate b over I , the other in coordinate d over J .

Which terms have nonzero expectation. In each such term the two increments lie in one coordinate from $\{a, b\}$, call it i , and one coordinate from $\{c, d\}$, call it i' . By cross-coordinate independence,

$$\mathbb{E}[(\varepsilon_{\alpha,i} - \varepsilon_{\alpha',i})(\varepsilon_{\gamma,i'} - \varepsilon_{\gamma',i'})] = 0 \quad \text{whenever } i \neq i', \quad (5)$$

so a term has nonzero expectation only when $i = i'$, that is, only when its two increments lie in the same coordinate. Writing that coordinate as i ,

$$\mathbb{E}[(\varepsilon_{\alpha,i} - \varepsilon_{\alpha',i})(\varepsilon_{\gamma,i} - \varepsilon_{\gamma',i})] = \sigma_i^2 g_{IJ}, \quad (6)$$

with the same coefficient $g_{IJ} = \mathbf{1}\{\alpha = \gamma\} - \mathbf{1}\{\alpha = \gamma'\} - \mathbf{1}\{\alpha' = \gamma\} + \mathbf{1}\{\alpha' = \gamma'\}$ as for the quadratic part. Such a common coordinate exists only if $\{a, b\}$ and $\{c, d\}$ share a coordinate, which gives the three cases below.

■ **(i) Disjoint coordinate pairs.**

If $\{a, b\} \cap \{c, d\} = \emptyset$, then the noise increment of any $L_{t,(a,b)}$ term (in a or b) and the noise increment of any $L_{t+m,(c,d)}$ term (in c or d) lie in different coordinates. By (5), every term of $L_{t,(a,b)} L_{t+m,(c,d)}$ has expectation 0, and thus $\text{Cov}(L_{t,(a,b)}, L_{t+m,(c,d)}) = 0$.

■ **(ii) One shared coordinate.**

If $\{a, b\} \cap \{c, d\} = \{j\}$, write $\{a, b\} = \{j, r\}$ and $\{c, d\} = \{j, s\}$, so r and s are the partners of j in (a, b) and (c, d) , respectively. A term of the product $L_{t,(a,b)} L_{t+m,(c,d)}$ has nonzero expectation only when both its increments lie in the shared coordinate j , namely the terms carrying the mean jump in the partner r (from $L_{t,(a,b)}$) and the mean jump in the partner s (from $L_{t+m,(c,d)}$). Collecting them, the contributing part of the product factors as

$$\left(\frac{1}{2} \sum_I s_I (\theta_{\alpha,r} - \theta_{\alpha',r})(\varepsilon_{\alpha,j} - \varepsilon_{\alpha',j})\right) \left(\frac{1}{2} \sum_J s_J (\theta_{\gamma,s} - \theta_{\gamma',s})(\varepsilon_{\gamma,j} - \varepsilon_{\gamma',j})\right), \quad (7)$$

the mean-jump-in- r part of $L_{t,(a,b)}$ times the mean-jump-in- s part of $L_{t+m,(c,d)}$, with $I = (\alpha, \alpha')$ varying over the first window's time pairs and $J = (\gamma, \gamma')$ varying over the second window's time pairs.

The sum in the first factor of (7) runs over the three time pairs of $L_{t,(a,b)}$, $\{(t, t+1), (t+1, t+2), (t, t+2)\}$, with $s_I = +1, +1, -1$ respectively. On the first two time pairs the mean differences

of r are single-gap jumps, δ_r^t across $(t, t+1)$ and δ_r^{t+1} across $(t+1, t+2)$. Only on $(t, t+2)$ does the mean difference split, as $\theta_{t,r} - \theta_{t+2,r} = \delta_r^t + \delta_r^{t+1}$. Substituting, the first factor becomes

$$\frac{1}{2} [\delta_r^t (\varepsilon_{t+2,j} - \varepsilon_{t+1,j}) + \delta_r^{t+1} (\varepsilon_{t+1,j} - \varepsilon_{t,j})].$$

The second factor is the same over $L_{t+m,(c,d)}$'s time pairs $\{(t+m, t+m+1), (t+m+1, t+m+2), (t+m, t+m+2)\}$, with partner s :

$$\frac{1}{2} [\delta_s^{t+m} (\varepsilon_{t+m+2,j} - \varepsilon_{t+m+1,j}) + \delta_s^{t+m+1} (\varepsilon_{t+m+1,j} - \varepsilon_{t+m,j})].$$

Multiplying the two, (7) is the sum of four terms,

$$\begin{aligned} & \frac{1}{4} \left[\delta_r^t \delta_s^{t+m} (\varepsilon_{t+2,j} - \varepsilon_{t+1,j}) (\varepsilon_{t+m+2,j} - \varepsilon_{t+m+1,j}) \right. \\ & + \delta_r^t \delta_s^{t+m+1} (\varepsilon_{t+2,j} - \varepsilon_{t+1,j}) (\varepsilon_{t+m+1,j} - \varepsilon_{t+m,j}) \\ & + \delta_r^{t+1} \delta_s^{t+m} (\varepsilon_{t+1,j} - \varepsilon_{t,j}) (\varepsilon_{t+m+2,j} - \varepsilon_{t+m+1,j}) \\ & \left. + \delta_r^{t+1} \delta_s^{t+m+1} (\varepsilon_{t+1,j} - \varepsilon_{t,j}) (\varepsilon_{t+m+1,j} - \varepsilon_{t+m,j}) \right]. \end{aligned}$$

■ (iii) **Equal coordinate pairs, $(a, b) = (c, d)$.**

Since $(a, b) = (c, d)$, a term of the product $L_{t,(a,b)} L_{t+m,(a,b)}$ has nonzero expectation when both its increments lie in coordinate a , or both lie in coordinate b . The contributing part of the product is then

$$\begin{aligned} & \left(\frac{1}{2} \sum_I s_I (\theta_{\alpha,b} - \theta_{\alpha',b}) (\varepsilon_{\alpha,a} - \varepsilon_{\alpha',a}) \right) \left(\frac{1}{2} \sum_J s_J (\theta_{\gamma,b} - \theta_{\gamma',b}) (\varepsilon_{\gamma,a} - \varepsilon_{\gamma',a}) \right) \\ & + \left(\frac{1}{2} \sum_I s_I (\theta_{\alpha,a} - \theta_{\alpha',a}) (\varepsilon_{\alpha,b} - \varepsilon_{\alpha',b}) \right) \left(\frac{1}{2} \sum_J s_J (\theta_{\gamma,a} - \theta_{\gamma',a}) (\varepsilon_{\gamma,b} - \varepsilon_{\gamma',b}) \right), \end{aligned} \quad (8)$$

whose first term is the product of the mean-jump-in- b parts of $L_{t,(a,b)}$ and $L_{t+m,(a,b)}$ (increments in a), and whose second term is the product of their mean-jump-in- a parts (increments in b). Each factor simplifies as in (7), so multiplying the two factors of the first term, the a -product (increments in a , jumps in b), gives four terms,

$$\begin{aligned} & \frac{1}{4} \left[\delta_b^t \delta_b^{t+m} (\varepsilon_{t+2,a} - \varepsilon_{t+1,a}) (\varepsilon_{t+m+2,a} - \varepsilon_{t+m+1,a}) \right. \\ & + \delta_b^t \delta_b^{t+m+1} (\varepsilon_{t+2,a} - \varepsilon_{t+1,a}) (\varepsilon_{t+m+1,a} - \varepsilon_{t+m,a}) \\ & + \delta_b^{t+1} \delta_b^{t+m} (\varepsilon_{t+1,a} - \varepsilon_{t,a}) (\varepsilon_{t+m+2,a} - \varepsilon_{t+m+1,a}) \\ & \left. + \delta_b^{t+1} \delta_b^{t+m+1} (\varepsilon_{t+1,a} - \varepsilon_{t,a}) (\varepsilon_{t+m+1,a} - \varepsilon_{t+m,a}) \right]. \end{aligned}$$

The second term, the b -product, is these four terms with a and b interchanged.

Simplifying the contributing part. Apart from the disjoint coordinate pairs case, which contributes 0, two cases remain: one shared coordinate, with contributing part (7), and equal coordinate pairs, with contributing part (8). In both of these cases, the covariance of the linear parts is the expectation of the contributing part, evaluated term by term from the expansions above: four terms for (7), and eight for (8) (four from each of its two products).

The following two facts hold in both cases and settle every occurrence of the change points.

- **Surviving term:** Each term of the expansions above is $\frac{1}{4} \delta_x^\tau \delta_y^{\tau'}$ times two noise increments. The mean jump δ_x^τ is taken across a gap $(\tau, \tau + 1)$ of the first window and the mean jump $\delta_y^{\tau'}$ across a gap $(\tau', \tau' + 1)$ of the second window (the windows on which $L_{t,(a,b)}$ and $L_{t+m,(c,d)}$ are built). Each window's noise increment lies over its other gap (each window has two gaps). By Assumption 1 no two change points fall in adjacent gaps. Since a mean jump across a gap without a change point is 0 and the two gaps of a window are adjacent, at most one mean jump over each window's two gaps is nonzero, and δ_x^τ and $\delta_y^{\tau'}$ with $\tau \neq \tau'$ can both be nonzero only when $|\tau - \tau'| > 1$. Hence each admissible occurrence of the change points **leaves at most one surviving term in each of the three four-term expansions above**, namely the one of (7) and the two of (8). If a window has no change point, every term of the three expansions has a mean jump from that window equal to 0, so every term vanishes and the covariance is 0.
- **Increment covariance:** The two noise increments of a term lie in the same coordinate, each over a gap. Writing I and J for these gaps (so I and J are pairs of consecutive times, not arbitrary time pairs), (6) gives their covariance as g_{IJ} times that coordinate's variance, where $g_{IJ} = 2$ if they are the same increment ($I = J$), $g_{IJ} = -1$ if I and J share exactly one time (adjacent gaps), and $g_{IJ} = 0$ if they share no time.

In the two cases below, a change-point occurrence that contributes is shown in **green** and one giving 0 in **red**.

■ One shared coordinate.

In the four-term expansion of (7), every term has its two mean jumps in the partners r and s and its two noise increments in the shared coordinate j . We sort the change-point occurrences by the number of change points.

- **No change point.** Every mean jump vanishes, so every term of the expansion of (7) is 0 and thus the covariance is 0.
- **One change point.** This contributes only at lag 0.
 - **Lag 0.** $L_{t,(a,b)}$ and $L_{t,(c,d)}$ are built on the single window $\{t, t + 1, t + 2\}$, and the single change point may lie in $(t, t + 1)$ or $(t + 1, t + 2)$. If it lies in $(t, t + 1)$, then $\delta_r^{t+1} = \delta_s^{t+1} = 0$, and the surviving term is $\frac{1}{4} \delta_r^t \delta_s^t (\varepsilon_{t+2,j} - \varepsilon_{t+1,j})^2$, a squared increment ($g_{IJ} = 2$), so $\mathbb{E}[L_{t,(a,b)} L_{t,(c,d)}] = \frac{1}{2} \sigma_j^2 \delta_r^t \delta_s^t$. A change point in $(t + 1, t + 2)$ is the symmetric case ($\delta_r^t = \delta_s^t = 0$), giving $\frac{1}{2} \sigma_j^2 \delta_r^{t+1} \delta_s^{t+1}$. In either case the lag-0 covariance equals $\frac{1}{2} \sigma_j^2 (\delta_r^t \delta_s^t + \delta_r^{t+1} \delta_s^{t+1})$ (since the summand for the change-point-free gap is 0).
 - **Lag 1.** A change point in $(t, t + 1)$ or $(t + 2, t + 3)$ lies in one window only, so every term of the expansion of (7) vanishes, since its mean jump from the other window is 0. A change point in the shared gap $(t + 1, t + 2)$ lies in both windows, but the surviving term's increments lie over $(t, t + 1)$ and $(t + 2, t + 3)$, which share no time ($g_{IJ} = 0$). Hence one change point **contributes nothing** at lag 1.
 - **Lag 2.** The windows $\{t, t + 1, t + 2\}$ and $\{t + 2, t + 3, t + 4\}$ share no gap, so the single change point lies in only one of them and every term of the expansion of (7) vanishes, since its mean jump from the other window is 0. Hence one change point **contributes nothing** at lag 2.
- **Two change points, one in each window.**

- **Lag 0.** $L_{t,(a,b)}$ and $L_{t,(c,d)}$ share the single window $\{t, t+1, t+2\}$, which holds at most one change point, so two change points cannot occur and this case is empty.
- **Lag 1.** The occurrences $\{(t, t+1), (t+1, t+2)\}$ and $\{(t+1, t+2), (t+2, t+3)\}$ are adjacent and excluded, leaving $\{(t, t+1), (t+2, t+3)\}$. Its surviving term is $\frac{1}{4}\delta_r^t \delta_s^{t+2} (\varepsilon_{t+2,j} - \varepsilon_{t+1,j})^2$, a squared increment ($g_{IJ} = 2$), giving $\frac{1}{2}\sigma_j^2 \delta_r^t \delta_s^{t+2}$.
- **Lag 2.** Only $\{(t, t+1), (t+3, t+4)\}$ contributes. The surviving term's increments lie over the adjacent gaps $(t+1, t+2)$ and $(t+2, t+3)$, which share only the time $t+2$, so $g_{IJ} = -1$, giving $-\frac{1}{4}\sigma_j^2 \delta_r^t \delta_s^{t+3}$. The pairs $\{(t, t+1), (t+2, t+3)\}$ and $\{(t+1, t+2), (t+3, t+4)\}$ give surviving terms whose increments (both in j) lie over gaps sharing no time, and $\{(t+1, t+2), (t+2, t+3)\}$ is adjacent and excluded.

Collecting these, the covariance of the linear parts is

$$\begin{aligned} \text{lag 0} &: \frac{1}{2}\sigma_j^2 (\delta_r^t \delta_s^t + \delta_r^{t+1} \delta_s^{t+1}), \\ \text{lag 1} &: \frac{1}{2}\sigma_j^2 \delta_r^t \delta_s^{t+2}, \\ \text{lag 2} &: -\frac{1}{4}\sigma_j^2 \delta_r^t \delta_s^{t+3}. \end{aligned}$$

■ Equal coordinate pairs.

By (8), the contributing part of $L_{t,(a,b)} L_{t+m,(a,b)}$ is a sum of two products, according to the coordinate in which both of a term's two noise increments lie: the *a-product*, from the terms with both increments in a (both mean jumps in b), and the *b-product*, from the terms with both increments in b (both mean jumps in a). Each is a one-shared product: the *a-product* is (7) with shared coordinate a and both partners b , and the *b-product* is the swap.

Take the *a-product*. We sort the change-point occurrences by the number of change points.

- **No change point.** Every mean jump vanishes, so every term of the *a-product*'s expansion is 0 and thus the *a-product* is 0.
- **One change point.** This contributes only at lag 0.
 - **Lag 0.** The product is $L_{t,(a,b)} L_{t,(a,b)}$, built on the single window $\{t, t+1, t+2\}$, and the single change point may lie in $(t, t+1)$ or $(t+1, t+2)$.
 - **Lags 1 and 2.** A single change point **contributes nothing**, exactly as in the one-shared case, including the shared-gap occurrence at lag 1.
- **Two change points,** one in each window, with the same occurrences as in the one-shared case.
 - **Lag 0.** The product is built on the single window $\{t, t+1, t+2\}$, which holds at most one change point, so two change points cannot occur and this case is empty.
 - **Lag 1.** The occurrence $\{(t, t+1), (t+2, t+3)\}$ contributes. The other occurrences are adjacent and excluded.
 - **Lag 2.** The occurrence $\{(t, t+1), (t+3, t+4)\}$ contributes. Every other occurrence gives 0, exactly as in the one-shared case, from gaps sharing no time or adjacent gaps.

Each product is evaluated as in the one-shared case, the *a-product* with σ_a^2 and the mean jumps of b , the *b-product* with σ_b^2 and the mean jumps of a . Adding the two, the covariance of the linear

parts for equal coordinate pairs is

$$\begin{aligned}\text{lag } 0 &: \frac{1}{2}\sigma_a^2((\delta_b^t)^2 + (\delta_b^{t+1})^2) + \frac{1}{2}\sigma_b^2((\delta_a^t)^2 + (\delta_a^{t+1})^2), \\ \text{lag } 1 &: \frac{1}{2}\sigma_a^2\delta_b^t\delta_b^{t+2} + \frac{1}{2}\sigma_b^2\delta_a^t\delta_a^{t+2}, \\ \text{lag } 2 &: -\frac{1}{4}\sigma_a^2\delta_b^t\delta_b^{t+3} - \frac{1}{4}\sigma_b^2\delta_a^t\delta_a^{t+3}.\end{aligned}$$

6 The assembled lagged autocovariance

Collecting the two parts at each lag gives the following.

Main result

Theorem 1 (Lag-0, lag-1 and lag-2 autocovariance). Assume the null model (Assumption 1), and take $a < b$, $c < d$.

Lag 0. When $(a, b) = (c, d)$,

$$\begin{aligned}\text{Cov}(\widehat{s}_{t,(a,b)}, \widehat{s}_{t,(a,b)}) &= \frac{5}{2}\sigma_a^2\sigma_b^2 + \frac{1}{2}\sigma_a^2((\delta_b^t)^2 + (\delta_b^{t+1})^2) \\ &\quad + \frac{1}{2}\sigma_b^2((\delta_a^t)^2 + (\delta_a^{t+1})^2);\end{aligned}$$

when $\{a, b\} \cap \{c, d\} = \{j\}$ with partners r, s ,

$$\text{Cov}(\widehat{s}_{t,(a,b)}, \widehat{s}_{t,(c,d)}) = \frac{1}{2}\sigma_j^2(\delta_r^t\delta_s^t + \delta_r^{t+1}\delta_s^{t+1});$$

and the covariance is 0 when the coordinate pairs are disjoint.

Lag 1. When $(a, b) = (c, d)$,

$$\begin{aligned}\text{Cov}(\widehat{s}_{t,(a,b)}, \widehat{s}_{t+1,(a,b)}) &= \frac{1}{2}\sigma_a^2\sigma_b^2 + \frac{1}{2}\sigma_a^2\delta_b^t\delta_b^{t+2} \\ &\quad + \frac{1}{2}\sigma_b^2\delta_a^t\delta_a^{t+2};\end{aligned}$$

when $\{a, b\} \cap \{c, d\} = \{j\}$ with partners r, s ,

$$\text{Cov}(\widehat{s}_{t,(a,b)}, \widehat{s}_{t+1,(c,d)}) = \frac{1}{2}\sigma_j^2\delta_r^t\delta_s^{t+2};$$

and the covariance is 0 when the coordinate pairs are disjoint.

Lag 2 (no quadratic part). When $(a, b) = (c, d)$,

$$\begin{aligned}\text{Cov}(\widehat{s}_{t,(a,b)}, \widehat{s}_{t+2,(a,b)}) &= -\frac{1}{4}\sigma_a^2\delta_b^t\delta_b^{t+3} \\ &\quad -\frac{1}{4}\sigma_b^2\delta_a^t\delta_a^{t+3};\end{aligned}$$

when $\{a, b\} \cap \{c, d\} = \{j\}$ with partners r, s ,

$$\text{Cov}(\widehat{s}_{t,(a,b)}, \widehat{s}_{t+2,(c,d)}) = -\frac{1}{4}\sigma_j^2\delta_r^t\delta_s^{t+3};$$

and 0 when the coordinate pairs are disjoint.

For equal coordinate pairs $(a, b) = (c, d)$, the lag-0 covariance is the quadratic part $\frac{5}{2}\sigma_a^2\sigma_b^2$ plus the linear-part squares. At lag 1 the quadratic part drops to $\frac{1}{2}\sigma_a^2\sigma_b^2$, and at lag 2 it is gone (one-dependent), leaving only the linear term. For a single shared coordinate only the linear term is present, at all three lags, and disjoint coordinate pairs give 0 throughout.